



**THE UNIVERSITY  
OF QUEENSLAND**  
AUSTRALIA

This exam paper must not be removed from the venue

Venue \_\_\_\_\_  
 Seat Number \_\_\_\_\_  
 Student Number 

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

  
 Family Name \_\_\_\_\_  
 First Name \_\_\_\_\_

**School of Mathematics & Physics**  
**Semester 2 Examinations, 2025**  
**MATH1061 Discrete Mathematics**

*This paper is for St Lucia Campus students.*

**Examination Duration:** 120 minutes

**Planning Time:** 10 minutes

**Exam Conditions:**

- Specified written or printed material permitted
- Casio FX82 series or UQ approved and labelled calculator only
- During Planning Time - Students are encouraged to review and plan responses to the exam questions

**Materials Permitted in the Exam Venue:**

***(Aside from approved calculators, no electronic aids are permitted e.g., laptops, phones)***

One A4 sheet of handwritten notes, double sided, is permitted.

**Materials to be supplied to Students:**

***Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.***

**Instructions to Students:**

***If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.***

Complete all examination questions in the space provided on this examination paper. If more space is required, use the backs of pages. The final page is a formula sheet, which you may detach. Nothing written on the formula sheet will be marked. Answers will be marked based on correctness, quality of expression, depth of understanding demonstrated, and clarity of explanation. If you have any queries about a particular examination question, or if you make any assumptions about the meaning of an examination question, please state this clearly at the start of your solution to the examination question.

**For Examiner Use Only**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| 12       |      |

Total \_\_\_\_\_

1. Let  $p$ ,  $q$ , and  $r$  be statement variables. Is the following argument valid? Justify your answer.

$$p \rightarrow q$$

$$\sim p \rightarrow r$$

$$\sim q \rightarrow r$$

$$\therefore q$$

**(3 marks)**

2. Write the negation of the following statement. Determine whether the original statement or the negation is true. Justify your answer.

$\forall x \in \mathbb{R}$ , if  $2x$  is irrational, then  $x$  is irrational.

**(4 marks)**

3. Use the Euclidean algorithm to compute  $\gcd(662, 414)$ .

**(3 marks)**

4. Let  $\{a_n\}_{n \geq 0}$  be the sequence defined by  $a_0 = 0$  and

$$a_{n+1} = a_n + 2n + 1, \quad n \geq 0.$$

- (a) Write down the values of  $a_1, a_2, a_3, a_4$ . (1 mark)

- (b) Based on your answers for Part (a), guess an explicit formula for this sequence.

(1 mark)

- (c) Prove that your guess is correct.

(3 marks)

5. Let  $A = \{0, 1, 2\}$  and  $B = \{-1, 0\}$ . Write down the following sets by listing their elements and using appropriate brackets.

(4 marks)

(a)  $A \cup B$ .

(b)  $A \cap B$ .

(c)  $\mathcal{P}(B)$ .

(d)  $A \times B$ .

6. Consider the function  $f : \mathbb{R} \rightarrow \mathbb{Z}$  defined by  $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor$ .

(a) Determine  $f(-1), f(0), f(1), f(2)$ . (2 marks)

(b) Is  $f$  one to one? Justify your answer. (2 marks)

(c) Is  $f$  onto? Justify your answer. (3 marks)

7. Let  $\mathbb{N}$  denote the set of positive integers. Define a relation on  $\mathbb{N} \times \mathbb{N}$  by

$$(a, b) \sim (c, d) \quad \text{if and only if} \quad a + d = b + c.$$

Prove that this is an equivalence relation.

**(5 marks)**

8. Solve the equation  $3x - 5 = 8$  in  $(\mathbb{Z}_{11}, +, \times)$ .

Hint: You may use the fact that  $3 \times 4 \equiv 1 \pmod{11}$ .

**(3 marks)**

**9.** Let  $E$  be the set of all even integers.

**(a)** Prove that  $|E| = |\mathbb{Z}|$ . **(5 marks)**

**Question 9 continues on the next page**

**9.(b)** Prove that  $E$  is a subgroup of  $(\mathbb{Z}, +)$ .

**(4 marks)**

- 10.** What is the coefficient of  $x^3$  in the expansion of  $(2x - 3)^6$ ? You should give your answer as a single integer, and show your working. **(2 marks)**

11. (a) How many distinct arrangements are there of the 10 letters in the word KOOKABURRA?

(Note: You do not need to compute an integer answer. You may give a mathematical expression that answers the question.)

(2 marks)

Question 11 continues on the next page

- 11.(b)** Suppose an arrangement is chosen randomly from the set of all arrangements in Part (a). What is the probability that in the chosen arrangement the two A's are not next to each other? Give the answer in percentage form.

**(3 marks)**

- 12.** Let  $n$  be a positive integer. Let  $K_n$  be the graph with  $n$  vertices such that each vertex is connected to every other vertex by a unique edge (but no vertex is connected to itself, i.e. there are no loops).

**(a)** Draw  $K_1$ ,  $K_2$ ,  $K_3$ ,  $K_4$ , and  $K_5$ . **(2 marks)**

**(b)** Write down the adjacency matrix of  $K_4$ . **(1 mark)**

**Question 12 continues on the next page**

**12.(c)** How many edges does  $K_n$  have? Justify your answer.

**(4 marks)**

**12.(d)** For which  $n$  does  $K_n$  have an Eulerian circuit? Justify your answer.

**(3 marks)**

**End of Examination**

## MATH1061/MATH7861 Examination Formula Page

**Logical Equivalences**

Given any statement variables  $p$ ,  $q$  and  $r$ , a tautology  $\mathbf{t}$  and a contradiction  $\mathbf{c}$ , the following logical equivalences hold.

|                                                                         |                                                             |                                                           |
|-------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------|
| <i>Commutative laws</i>                                                 | $p \wedge q \equiv q \wedge p$                              | $p \vee q \equiv q \vee p$                                |
| <i>Associative laws</i>                                                 | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$        | $(p \vee q) \vee r \equiv p \vee (q \vee r)$              |
| <i>Distributive laws</i>                                                | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |
| <i>Identity laws</i>                                                    | $p \wedge \mathbf{t} \equiv p$                              | $p \vee \mathbf{c} \equiv p$                              |
| <i>Negation laws</i>                                                    | $p \vee \sim p \equiv \mathbf{t}$                           | $p \wedge \sim p \equiv \mathbf{c}$                       |
| <i>Double negative laws</i>                                             | $\sim(\sim p) \equiv p$                                     |                                                           |
| <i>Idempotent laws</i>                                                  | $p \wedge p \equiv p$                                       | $p \vee p \equiv p$                                       |
| <i>Universal bound laws</i>                                             | $p \vee \mathbf{t} \equiv \mathbf{t}$                       | $p \wedge \mathbf{c} \equiv \mathbf{c}$                   |
| <i>De Morgan's laws</i>                                                 | $\sim(p \wedge q) \equiv \sim p \vee \sim q$                | $\sim(p \vee q) \equiv \sim p \wedge \sim q$              |
| <i>Absorption laws</i>                                                  | $p \vee (p \wedge q) \equiv p$                              | $p \wedge (p \vee q) \equiv p$                            |
| <i>Negations of <math>\mathbf{t}</math> and <math>\mathbf{c}</math></i> | $\sim \mathbf{t} \equiv \mathbf{c}$                         | $\sim \mathbf{c} \equiv \mathbf{t}$                       |

**Set Identities** For all sets  $A$ ,  $B$  and  $C$  that are subsets of a universal set  $U$ :

1. Commutative Laws (a)  $A \cup B = B \cup A$  (b)  $A \cap B = B \cap A$
2. Associative Laws (a)  $(A \cup B) \cup C = A \cup (B \cup C)$   
(b)  $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a)  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$   
(b)  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a)  $A \cup \emptyset = A$  (b)  $A \cap U = A$
5. Complement Laws (a)  $A \cup A^c = U$  (b)  $A \cap A^c = \emptyset$
6. Double Complement Law  $(A^c)^c = A$
7. Idempotent Laws (a)  $A \cup A = A$  (b)  $A \cap A = A$
8. Universal Bound Laws (a)  $A \cup U = U$  (b)  $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a)  $(A \cup B)^c = A^c \cap B^c$  (b)  $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a)  $A \cup (A \cap B) = A$  (b)  $A \cap (A \cup B) = A$
11. Complement of  $U$  and  $\emptyset$  (a)  $U^c = \emptyset$  (b)  $\emptyset^c = U$
12. Set Difference Law  $A - B = A \cap B^c$

**The Quotient-Remainder Theorem** Given any integer  $n$  and any positive integer  $d$ , there exist unique integers  $q$  and  $r$  such that  $n = dq + r$  and  $0 \leq r < d$ .

**Unique Factorisation Theorem** Given any integer  $n > 1$ , there exists a positive integer  $k$ , distinct prime numbers  $p_1, p_2, \dots, p_k$ , and positive integers  $e_1, e_2, \dots, e_k$  such that  $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$ , and any other expression for  $n$  as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

**Schröder-Bernstein Theorem** For all sets  $A$  and  $B$ , if  $|A| \leq |B|$  and  $|A| \geq |B|$  then  $|A| = |B|$ .

**Binomial Theorem** Given any real numbers  $a$  and  $b$ , and any nonnegative integer  $n$ ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$